

PARTICULAR SPECIFICATION P1

Overview of the Works

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P1.1 DEFINITIONS

- (a) “ACMV” refers to air conditioning and mechanical ventilation;
- (b) “BIM” refers to Building Information Management;
- (c) “C1A contractor” refers to the contractor responsible for Contract 1A – Site Development Works;
- (d) “C4A contractor” refers to the contractor responsible for Contract 4A – Biosolids and Digesters Works;
- (e) “C5A contractor” refers to the contractor responsible for Contract 5A – Plant Monitoring and Control Systems Works;
- (f) “Common Areas” refers to the areas of the TWRP Area that are managed by the C1A contractor and are not within another contractor’s site, through which the Contractor shall access the Site and in which the Contractor shall carry out certain work as required by the Drawings and Specifications;
- (g) “CSS” refers to Coarse Screen Shaft(s);
- (h) “DLM” refers to Domestic Liquids Modules;
- (i) “Drawings” refers to the Volume 5 Part 1 and 2 Tender Drawings and BIM Tender Model;
- (j) “DTSS” refers to the Deep Tunnel Sewerage System;
- (k) “DTSS1” refers to Deep Tunnel Sewerage System Phase 1 Project;
- (l) “DTSS2” refers to Deep Tunnel Sewerage System Phase 2 Project;
- (m) “DUW” refers to Domestic Used Water;
- (n) “ERSS” refers to Earth Retaining and Stabilising Structure;
- (o) “GTP” - Gas Transmission Pipeline refers to existing DN700 gas pipeline from Jurong island to TPG, part of which is routed along the coast of TVB;
- (p) “ILM” refers to Industrial Liquids Modules;
- (q) “IMC” refers to Instrumentation and Monitoring Contractor;
- (r) “IPS” refers to Influent Pumping Station(s);
- (s) “IUW” refers to Industrial Used Water;
- (t) “IW” refers to Industrial Water;

- (u) "IWMF" refers to NEA's Integrated Waste Management Facility, being constructed on TVB adjacent to the TWRP Area;
- (v) "MDF" refers to Main Distribution Frame;
- (w) "MEICA" refers to mechanical, engineering, instrumentation, control and automation;
- (x) "NEA" refers to the National Environment Agency;
- (y) "NEA contractors" refers to all contractors appointed by NEA including IWMF contractors;
- (z) "NW" refers to NEWater;
- (aa) "NSG" refers to North Staging Ground;
- (bb) "NSG Administration Facility" refers to the administration facility at NSG;
- (cc) "OCF" refers to Odour Control Facilities;
- (dd) "Other contractors" refers to any contractor appointed by the Board or NEA including the T-08 contractor;
- (ee) "RORO" refers to Roll-On Roll-Off;
- (ff) "Plant Water" refers to TWRP DLM MBR filtrate;
- (gg) "PUB" or "The Board" refers to the Public Utilities Board;
- (hh) "Site" refers to the Contractor's Staging Area and worksite within TWRP Area as shown on the Drawings;
- (ii) "Site Offices" refers to centralised Site Offices for the Board and S.O.;
- (jj) "site office" refers to the Contractor's site office;
- (kk) "SOP" refers to Schedule of Prices;
- (ll) "SSG1" refers to South Staging Ground 1 or South Staging Ground;
- (mm) "SSN" refers to Sidestream Nitrogen Removal or Sidestream N Removal;
- (nn) "TE" refers to Treated Effluent from Jurong Water Reclamation Plant or Ulu Pandan Water Reclamation Plant;
- (oo) "THP" refers to Thermal Hydrolysis Process;
- (pp) "TPLS" refers to Tuas Port Local Sewer;
- (qq) "TPG" refers to Tuas Power Generation Pte Ltd;

- (rr) “TVB” refers to Tuas View Basin, which includes;
 - (i) The Route 2 and Route 3 bridges connecting the Tuas View Basin with Tuas South Avenue 9 up to the limits of land owned by the Ministry of Sustainability and the Environment (MSE), including land lot numbers MK07-04791X, MK07-04155V and MK07-04216A;
 - 1) The TWRP Area, including the Site;
 - 2) The T-08 site; and
 - 3) The IWMF Area.
- (ss) “TWRP” refers to Tuas Water Reclamation Plant;
- (tt) “TWRP Area” refers to the area on TVB on which TWRP will be sited. This includes the Site including foreshore areas. The TWRP Area has the following land lot numbers:
 - (i) MK07-05035M; and
 - (ii) MK07-05037V.
- (uu) “TWRP Project” refers to TWRP Project Phase 1;
- (vv) “T-08” refers to Design and Construction of Sewer Tunnels for the Deep Tunnel Sewerage System Phase-2 Project Contract T-08;
- (ww) “WRN” refers to Water Reclamation Network (PUB);
- (xx) “WRP” refers to Water Reclamation Plant (PUB); and
- (yy) “WSN” refers to Water Supply Network (PUB).

P1.2 INTRODUCTION

- (a) The Board intends to engage a Contractor to provide the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities for TWRP, which will be developed as part of DTSS2.
- (b) This section of the Specifications gives an overview of the objectives of the project and an outline of the overall scope of work. Reference shall be made to other parts of the tender documents, which set out the detailed requirements for the Works and highlight the Contractor’s obligations.

P1.3 DESCRIPTION OF THE PROJECT

- (a) DTSS employs a conveyance system of link sewers and deep tunnel sewers to transport used water by gravity to centralise WRPs in coastal areas. Used water is treated and further purified into high-grade reclaimed water (NEWater) with any excess treated used water that is not used for process or other plant water needs, being discharged to the sea.

- (b) DTSS1, which serves the eastern part of Singapore, was completed in 2008. DTSS2 will see the existing deep tunnel system extended to serve the western part of Singapore. To facilitate used water treatment and the production of NEWater and Industrial Water, there will be two tunnels to convey IUW and DUW to TWRP separately.
- (c) TWRP will use membrane bioreactor (MBR) technology to treat IUW and DUW in separate treatment modules. IUW will be treated in a 150 MLD ILM and some will be further purified to meet IW standards. DUW will be treated in a distinct 650 MLD DLM and some will be further purified to produce NW.
- (d) Biosolids will be treated in a common facility to stabilise sludge and produce biogas. In addition, septage and greasy waste from commercial outlets will be imported by road tankers. The septage will be treated in the ILM and greasy waste will be pre-treated before being co-digested with used water sludge.
- (e) Overall, TWRP will employ advanced treatment processes to minimise specific energy consumption and sludge production and at the same time maximise water recovery for re-use and biogas production for the generation of electricity.
- (f) A first in Singapore, TWRP will be co-located with the IWMPF, to be developed by NEA, in one of the world's first major example of industrial ecology, wherein the by-products of one plant is used as a resource for a neighbouring facility. The integration of TWRP and IWMPF will, collectively known as the Tuas Nexus, will allow the Board and NEA to exploit the synergies associated with the water-energy-waste nexus. An example is the co-digestion of food waste with used water sludge at TWRP to increase the yield of biogas, which will be utilised at IWMPF to improve steam quality and give rise to higher overall plant thermal efficiency. This will in turn increase electricity production and enable IWMPF to export more energy to the grid while allowing both facilities to be energy self-sufficient. IWMPF will provide electricity for the daily operations of TWRP.
- (g) TWRP will be developed in phases. The initial phase of TWRP (i.e. TWRP Phase I) is envisaged to be completed in 2025. Future phases of TWRP will be constructed progressively thereafter. The anticipated contract packages referenced herein, are for the completion of TWRP Phase I and, where appropriate, include works which will facilitate the implementation of future expansions of TWRP.

P1.4 PRELIMINARY CONTRACT PACKAGES

- (a) The Board intends progressively to appoint contractors to carry out construction of TWRP (herein referred to as TWRP contractors). It is anticipated that TWRP will be delivered using seventeen (17) contracts as described in Table P1-1.

Table P1-1: Preliminary Contract Packaging for TWRP

Package	Description
Contract 1A	Site Development Works
Contract 2A	Influent Pumping Stations
Contract 2B	Product Water Storage and Pumping Station
Contract 3A	Industrial Liquids Modules – Civil and Structural Works
Contract 3B1	Industrial Liquids Module 1 – MEICA
Contract 3B2	Industrial Liquids Module 2 – MEICA
Contract 3B3	Industrial Water Plant
Contract 3C	Domestic Liquids Modules – Civil and Building Services Works
Contract 3D1	Domestic Liquids Module 1 – MEICA & NEWater
Contract 3D2	Domestic Liquids Module 2 – MEICA
Contract 3E	Liquids Modules Odour Control Facilities
Contract 4A	Biosolids and Digesters
Contract 4B	Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities (This Contract)
Contract 5A	Plant Monitoring and Control System
Contract 6A	Administration and Operations Buildings
Contract 6B	Site Completion Works
Contract 6C	Wet Weather Treatment Facility

- (b) The list of contracts may be refined at the Board’s discretion.

P1.5 OVERALL PROGRAM FOR TWRP

- (a) TWRP will be developed in phases to cater for projected used water flows and loads. The current project, planned to be operational in 2025, is considered as Phase 1 with additional phases to be undertaken up to the planned ultimate capacity of the plant. The Intermediate milestone are as shown in Table P1-2. Other key project milestones are provided in Particular Specification P2 Preliminaries and General.

Table P1-2: TWRP Intermediate Milestones for Phase 1

Milestone	Description
MD1	Industrial Water into Supply from Domestic Liquids Module 1
MD2	NEWater into Supply from Domestic Liquids Module 1
MD3	Completion of Phase 1 for Used Water Treatment

- (b) Upon successful completion of the whole of the Works to the satisfaction of the S.O.:
 - (i) the Contractor will receive Substantial Completion for the whole of the Works; and
 - (ii) the Contractor shall maintain and operate the Works in collaboration with the S.O. to facilitate operation of the TWRP.
- (c) The duration and activities to be provided by the Contractor during the Operations and Maintenance Period (OM) is set out in Particular Specification P12 - Operations and Maintenance.

P1.6 AUTHORITY OF SUPERINTENDING OFFICER (S.O.) AND S.O. REPRESENTATIVE

- (a) The S.O. and his appointed representative are appointed by the Board for the purposes of Part 1 – Design and Build Works.

P1.7 LOCATION OF THE SITE AND ACCESSIBILITY

- (a) The Site is located within the TWRP Area of TVB, which is in the west of Singapore. TVB is bounded by the existing TPG industrial facility to the north and the TPG seawater intake channel to the west. To the east and south, TVB fronts the Straits of Singapore.
- (b) TVB and the Site can be approached from Tuas South Avenue 9. The Site location and access is shown in the Drawings.
- (c) The T-08 site is located immediately to the north of the Site.
- (d) Two elevated access bridges provide access to TVB, namely:
 - (i) Route 2 - from Tuas South Lane; and
 - (ii) Route 3 - from Tuas South Avenue 12.
- (e) For Contract 4B, access to the TWRP Area and the Site will normally be via Route 2 and/or Route 3. There are other access points to the TWRP Area and the Site that may be used on an as needed or emergency basis including, road access through TPG (for emergency use only), and RORO Landing Facility. These are shown on the Drawings.

- (f) During the Contract Period, access to the TWRP Area will be controlled by the C1A contractor, who will also be responsible, for the overall management, security, health, safety and environment of the TWRP Area as well as maintenance of Routes 2 and 3 leading to TVB. The C1A contractor will also manage and maintain the RORO Landing Facility. The Contractor shall adhere to the C1A contractor's requirements for access, security, health, safety and environment of the TWRP Area, where applicable, unless otherwise instructed by the S.O.. At times, access through Route 2 or Route 3 may be restricted. However, at least one will be available for access at all times barring unforeseen emergency situations.

P1.8 THE SITE

- (a) The Site is shown on the Drawings and comprises:
 - (i) Working areas comprising the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities;
 - (ii) External working areas for sitewide pipe rack, storm drainage and electrical & ICA manholes and duct banks; and
 - (iii) Designated staging area and any temporary working areas as described herein or that may be made available by the S.O. during the Contract.
- (b) Possession of the Site, or parts thereof, to be given by the Board will be subject to any restrictions highlighted in the Contract.
- (c) The Contractor shall work with the S.O. to co-ordinate his work with Other contractors who require access to the Site and shall share Site access.
- (d) The Contractor shall note the progressive transfer of work areas within the Site as detailed in Particular Specification P2 Preliminaries and General. The Contractor shall work with the Other contractors and the S.O. for access to the transferred work areas within the Site.
- (e) There will be times when work will be required within Common Areas. The Contractor shall liaise with the S.O. and the Other contracts to coordinate and agree access to these areas to perform such work.
- (f) The Contractor shall note that Other contractors will be carrying out their works within their works boundary including the common areas. These include excavations, foundations, sub structures, super structures, utilities, piping works, etc. The construction activities of this Contract and other TWRP contracts will run concurrently. The Contractor shall liaise with the S.O. and Other contractors to coordinate and agree access to these areas to perform such work.
- (g) The Contractor shall co-operate with the Board and the S.O. to manage and facilitate all of its interfaces with the various contracts.

P1.9 OVERVIEW OF SCOPE OF CONTRACT 4B

P1.9.1 Introduction

- (a) This section gives an outline of the overall scope of work. Reference should be made to other parts of the Specifications, which set out the detailed requirements.
- (b) The major Design and Build works to be provided under the Contract include:
 - (i) Digestion Pre-treatment Facility;
 - (ii) Sidestream Nitrogen Removal Facility;
 - (iii) Sitewide pipe rack and interconnecting pipework between Biosolids area and ILM;
 - (iv) Interfaces between the facilities and Sitewide, ILM 1 facility, Biosolids facility and IWMF.
- (c) The Contract includes major process units, as listed above, and this includes the integration of the units and the mitigation of any impact, direct or indirect, that one has on the other, this includes any inhibitory effects that the THP may have on the influent to the SSN facility.
- (d) The Contract includes a period of Operations and Maintenance (OM3) of the Digestion Pre-treatment and Sidestream Nitrogen Removal Facilities after construction of TWRP and described further in Particular Specification P12 Operations and Maintenance.

P1.9.2 Coordination with C1A contractor

- (a) The C1A contractor is responsible for the overall management, maintenance and coordination of activities within the TWRP Area under authority of the S.O. and subject to the S.O.'s approval. The Contractor will be required to provide details of daily activities and coordinate with the C1A contractor to ensure the smooth operation of the TWRP project which shall include activities as described in Particular Specification P2 Preliminaries and General.

P1.9.3 Construction Support Resources and Infrastructure

- (a) The following infrastructure shall be provided by the Contractor to support the construction of the Works:
 - (i) Connection of Site utilities to the Contractor's allocated Staging Ground to support construction from termination points provided under Contract 1A for potable water, NEWater, electricity, telecommunications distribution network as appropriate and shown on the Drawings;

- (ii) Earthworks – excavation, backfill, fill, compaction, transportation, disposal of excavation material to approved disposal site, provisions of required fill material to Site;
- (iii) Temporary drainage infrastructure (including pumping and pipework) within the Site to discharge to the TWRP Area storm water drainage system (temporary and permanent) provided under Contract 1A; and
- (iv) Temporary sanitation infrastructure within the TWRP Area (including pipework and pumping to PUB network) provided under Contract 1A.

P1.9.4 BIM Delivery

- (a) The Works shall be delivered using digital engineering or building information management (BIM). Detailed BIM requirements in Appendix P2C – Employer Information Requirements and General Specification G1.2 Information Management.
- (b) The application of BIM for the Works shall include the following as a minimum:
 - (i) All designs by the Contractor shall be carried out using BIM;
 - (ii) Permanent and temporary works shall be modelled;
 - (iii) Whenever 2D construction drawings are required, these shall be generated and extracted from the BIM model;
 - (iv) The output of BIM models shall be linked to the approved Programme for the Works using Primavera P6 software to produce construction sequence simulations (4D) for all temporary and permanent facilities;
 - (v) The output of 3D and 4D models shall be used to produce time sequenced models of the Board's contract expenditure (5D);
 - (vi) Construction operations building information exchange (COBie) data shall be provided for all modelled assets (6D); and
 - (vii) Select Cobie data shall be provided for use in the Board's asset management system (AMS).
- (c) It is critical to the Board and the S.O. that BIM is employed as specified to deliver the Works.
- (d) Schedule No. 1 (Volume 4 – Schedule of Prices and Fixed Schedule of Rates) includes a lump sum for BIM Delivery. Refer to the Preamble to Schedule of Prices for payment details. The Contractor shall comply with the specified BIM requirements given in Appendix P2C – Employer Requirements and General Specification G1.2 Information Management.

P1.9.5 Other Activities

- (a) The Contractor shall undertake other activities required by the Contract, including but not limited to the following:
 - (i) Additional data and information acquisition;
 - (ii) Agencies compliance and liaisons;
 - (iii) Comply with all statutory requirements imposed whether stated in the Tender documents or imposed later as a requirement of statutory approval by any party. The Contractor is to note that all fees and charges imposed by Statutory Boards and Government Authorities pertaining to all necessary applications / processing / submissions etc. shall be borne by the Contractor and these shall be deemed to have been included in his sum/prices;
 - (iv) Provide all necessary protection to existing properties, boundary wall, road elements, underground services during the entire duration of Works. Protection and provision of temporary support to any detected utilities and services which are not to be diverted under the Contract; The Contractor shall note that some works are in close proximity to the Tuas Port Local Sewer constructed under Contract 1A. The Contractor can obtain the information from the S.O. upon commencement and shall consider these existing assets in its works;
 - (v) Provide to the S.O. in a timely manner all necessary information related to the Works required to enable the S.O. to apply for and obtain BCA Green Mark certification. The Contractor shall note that the cost of providing such information shall be deemed to be included in his tendered sum;
 - (vi) Post-condition surveys of the Site and areas within the influence zone of the Works;
 - (vii) Pre-construction surveys;
 - (viii) Post construction surveys;
 - (ix) Record the progress of the Works;
 - (x) Provide maintenance services for the installed equipment during and after Contract award for the specified period;
 - (xi) Provide as-built and operation and maintenance information; and
 - (xii) Rectify defects during the Defects Liability Period (DLP).

P1.10 CONTRACT EXECUTION REQUIREMENTS

- (a) The Contractor shall plan the Works to a great level of detail and prioritise the completion of all permanent and temporary work to meet the contractual milestone dates. This is critical to facilitate the construction activities within the Site and the Contractor shall work closely with the S.O. to meet this target.
- (b) The Contractor shall cooperate and work closely with the S.O. and TWRP contractors in the planning of activities. Ample notice shall be provided to the S.O. in the event of any work activity under the Contract affecting TWRP contractors (such as road closures, cranes operations, major deliveries, etc.). In the event of any conflicting activities occurring at the same area or same time, the S.O. will make the final decision in prioritising activities to be carried out. The Contractor shall comply with the S.O.'s direction in this regard.

P1.11 WORKING WITH THE S.O. AND OTHER PARTIES

- (a) During the Contract period, the Contractor shall be required to interface extensively with the S.O. and take part in regular planning discussions with:
 - (i) All TWRP contractors (listed herein or yet to be identified);
 - (ii) The nominated IMC appointed by the Board;
 - (iii) Any contractor appointed by NEA for the delivery of the IWMF;
 - (iv) Officers from the Board;
 - (v) Agencies; and
 - (vi) Any contractor appointed by the Board.
- (b) Co-ordination between Other contractors will be paramount if the project is to meet the Board's objectives. The Contractor shall work with the S.O. to ensure the timely flow of technical and project execution information between the Contractor and the Other contractors and third parties.

P1.12 PROGRAMME FOR THE WORKS

- (a) The Time for Completion date(s) are set out in the Appendix to the PSSCOC.
- (b) In addition to the Time for Completion milestones, the Contractor shall programme the Works to meet the key project milestones given in Particular Specification P2 Preliminaries and General.

END OF SECTION